29.**Solution:**

Volume of $KMnO_4$ solution used ∞ amount of H_2O_2 present. Hence if the given reaction is of the first order, it must obey the equation

$$k = \frac{2.303}{t} \log \frac{a}{(a-x)}$$

$$k = \left(\frac{2.303}{t}\right) \log \left(\frac{V_0}{V_t}\right)$$

In the present case, V_o = 46.1 ml.

The value of *k* at each instant can be calculated as follows:

t (*min*)	V_t	
		$k = \frac{46.1}{\text{---}} \log \frac{46.1}{29.8}$
10	29.8	$k = \frac{46.1}{\text{---}} log \frac{46.1}{29.8} = 0.0436\ min^{-1}$
20	19.3	$k = \frac{46.1}{\text{---}} log \frac{46.1}{19.3} = 0.0435\ min^{-1}$

Thus, the value of *k* comes out to be nearly constant. Hence it is a reaction of the first order.

30. Solution:

i) For the first order reaction $k = \frac{2.303}{t} \log \frac{a}{a - x}$

Assume, a = 100 %, x = 40%, t = 50 minutes

Therefore, a - x = 100 – 40 = 60

k = (2.303 / 50) log (100 / 60)

k = 0.010216 min^{-1}

Hence the value of the rate constant is 0.010216 min^{-1}

ii) t = ?, when x = 80%,

Therefore, a - x = 100 – 80 = 20

From above, k = 0.010216 min^{-1}

t = (2.303 / 0.010216) log (100 / 20)

t = 157.58 min

The time at which the reaction will be 80% complete is 157.58 min.

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